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Course: Digital Image Processing

A comparative study of RAG techniques for generating image captions

PHASE 1 REPORT — Data exploration and pre-processing

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Part I

Data Exploration and Pre-processing

1.1 Introduction

Automatic image captioning is a computer vision task that aims to generate natural-language descriptions of images. This project studies how different Retrieval-Augmented Generation (RAG) techniques influence the quality, relevance, and visual grounding of generated image captions. RAG combines information retrieval with text generation. Instead of relying only on the knowledge learned during training, it retrieves relevant information from an external knowledge base to support caption generation.

1.2 Problem Statement

The project investigates how RAG strategies can support image captioning and how retrieval quality influences generated captions. The assigned group studies the Self-RAG technique, within a shared comparative-study protocol.

1.3 Objectives of Phase 1

The main objective of this phase is to explore, process, and prepare the MS COCO 2017 dataset for the implementation of the RAG techniques.

This phase covers several activities:

- Exploring the MS COCO 2017 dataset and understanding its structure.
- Analyzing object categories, scene complexity, and caption characteristics.
- Verifying the integrity of images and annotations.
- Preprocessing images and captions.
- Establishing reproducible knowledge-base, validation, and test splits.
- Organizing the dataset and its associated metadata.
- Generating image embeddings and preparing the vector index for retrieval experiments.

Overall, this phase establishes a structured and reliable data foundation for the implementation and comparison of different RAG techniques.

1.4 Overview of the MS COCO 2017 Dataset

1.4.1 Dataset Description

The project uses the Microsoft Common Objects in Context (MS COCO) 2017 dataset, which contains real-world images showing different objects, scenes, and interactions.

The dataset provides image annotations, object categories, and natural-language captions. In this project, two types of annotations are used:

Caption annotations: These provide natural-language descriptions of the images.

Instance annotations: These provide information about objects in the images, such as their categories and bounding boxes.

1.4.2 Dataset Sources and Files

The project uses the following annotation files:

- `captions_train2017.json`: Contains reference captions for training images.
- `instances_train2017.json`: Contains object-instance annotations for training images.
- `captions_val2017.json`: Contains reference captions for validation images used as the test set in this project.
- `instances_val2017.json`: Contains object-instance annotations for validation images, when needed for the analysis.

The image files are downloaded separately from the annotation files. Each image is linked to its annotations using its unique image ID.

1.5 Statistical Exploration of the Dataset

1.5.1 Object Category Analysis

Object Category Distribution

This analysis examines the frequency of object categories in the MS COCO Knowledge Base. Each object annotation is counted according to its category ID using Python's Counter function. The category IDs are then mapped to their corresponding names using the COCO API.

The results are sorted in descending order to identify the most frequent categories. The following bar chart presents the 50 categories with the highest annotation counts.

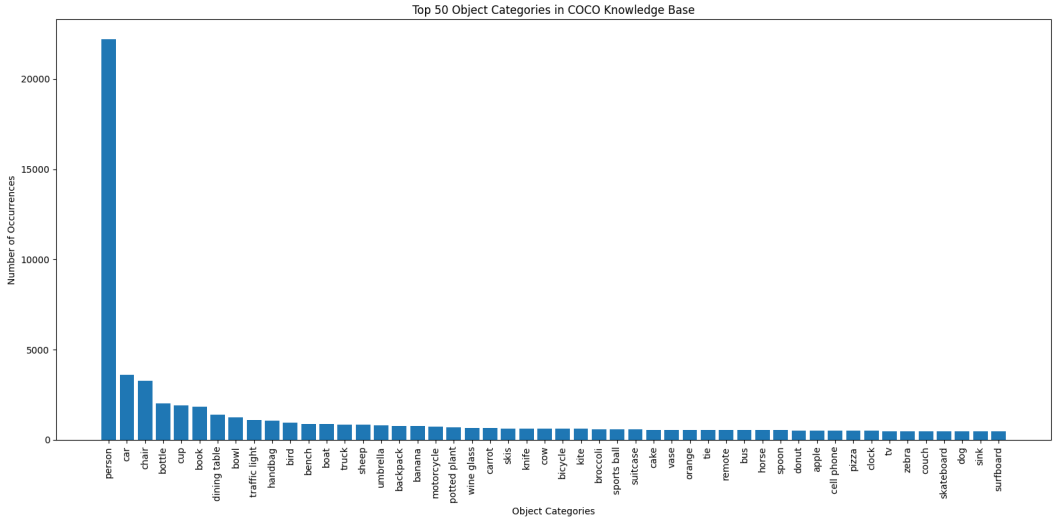


Figure 1.1: Top 50 Object Categories in the COCO Knowledge Base

Number of Images Containing Each Category

This analysis measures how many distinct images contain each object category in the Knowledge Base.

For each category, the corresponding image IDs are collected using a set to avoid counting the same image more than once. The number of unique image IDs is then calculated and added to the category statistics table.

This analysis provides additional information about the distribution of object categories across images, rather than simply counting the total number of object annotations.

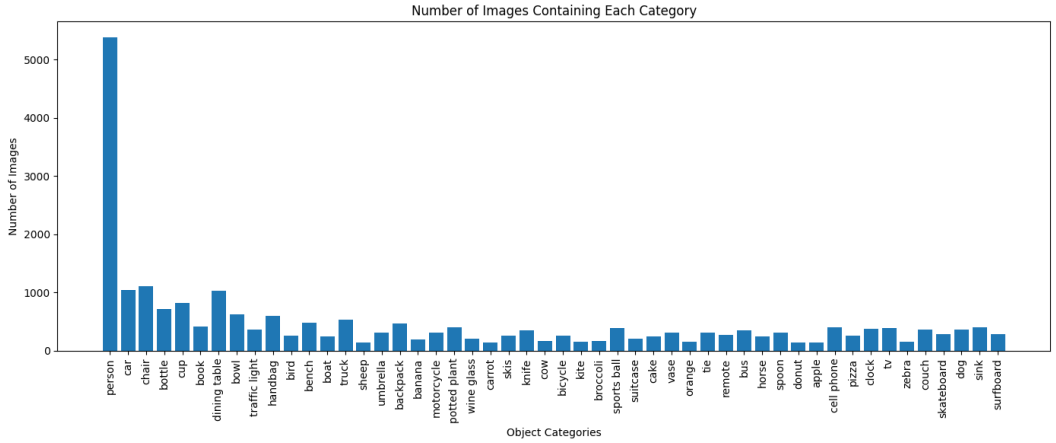


Figure 1.2: Number of Images Containing Each Category

1.5.2 Scene Complexity Analysis

Number of Objects per Image

This exploration aims to examine the number of objects present in each image of the MS COCO Knowledge Base. It provides an overview of how many objects typically appear in an image and helps identify differences in scene complexity.

Understanding the number of objects per image is important for this project because images with different numbers of objects may present different levels of difficulty for image captioning.

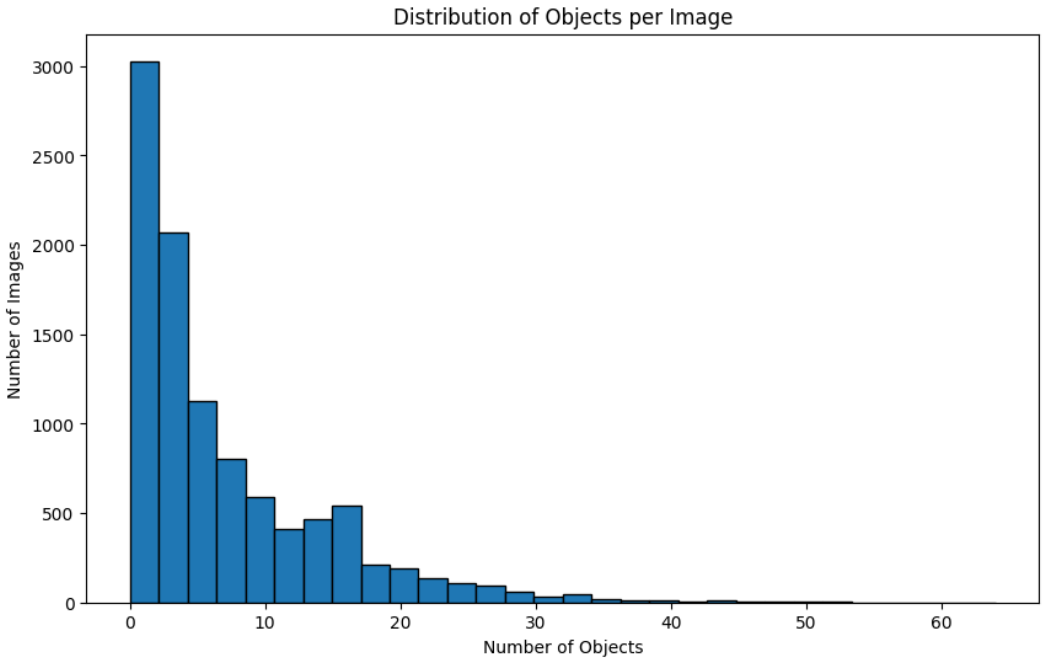


Figure 1.3: Number of Objects per Image

1.5.3 Scene Complexity Analysis

Scene Classification

Scene complexity classification consists of categorizing images into three levels (simple, medium, and complex) based on the number of objects present in each image. In this study, images containing up to 3 objects are classified as simple scenes, those containing between 4 and 9 objects are classified as medium scenes, and those containing 10 or more objects are classified as complex scenes. This classification provides an estimate of the visual complexity of each image and helps analyze the relationship between scene complexity and image captioning.

Scene Complexity	Number of Images
Simple Scenes	4264
Medium Scenes	3096
Complex Scenes	2640
Total	**10000**

Table 1.1: Distribution of Images According to Scene Complexity

1.5.4 Object Co-occurrence Analysis

Object Co-occurrence

The objective of this analysis is to identify the object categories that frequently appear together within the same images. It helps discover common co-occurrence patterns between objects and provides insights into the relationships between different object categories in the dataset.

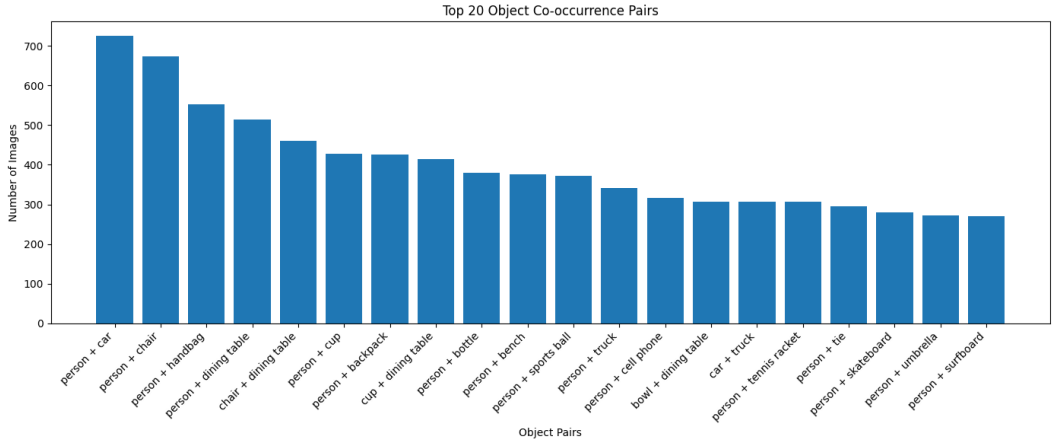


Figure 1.4: Object Co-occurrence

1.5.5 Caption Statistical Analysis

Caption Length Analysis

The objective of this analysis is to examine the length of image captions by measuring the number of words in each description. It helps understand how captions vary in length and whether the dataset mainly contains short, medium, or long descriptions.

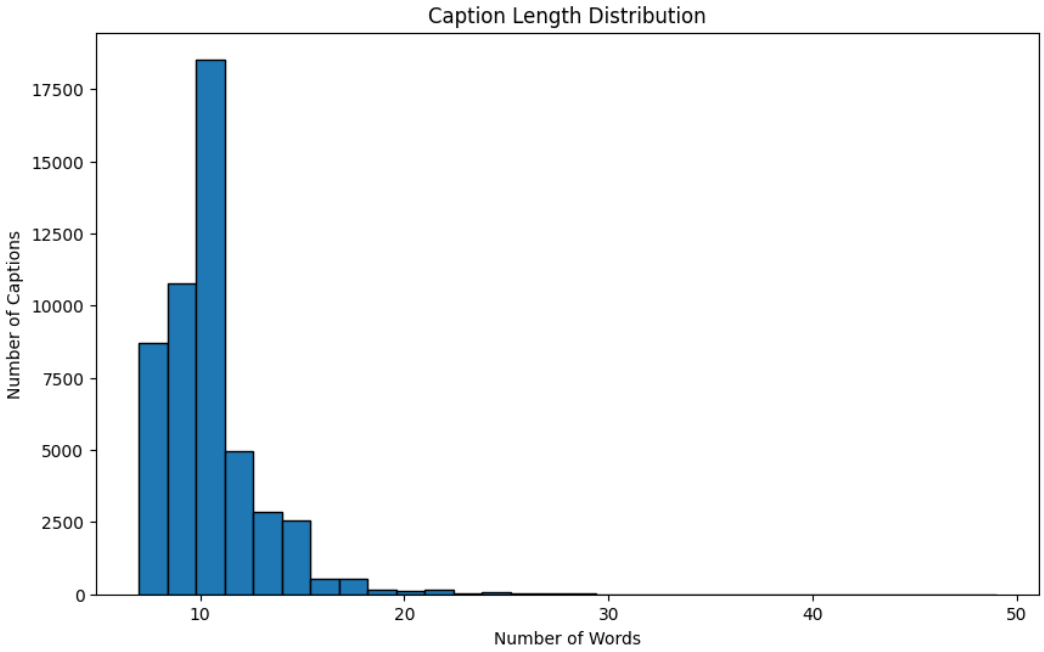


Figure 1.5: Caption Length Analysis

Vocabulary Size

The objective of this analysis is to measure the vocabulary diversity within the image captions. It examines the total number of word occurrences and the vocabulary size (number of unique words) to evaluate the lexical richness of the descriptions in the dataset.

Metric	**Value**
Total Word Occurrences	524,313
Vocabulary Size (Unique Words)	9,576

Table 1.2: Vocabulary Statistics of the Captions

Most Frequent Words

The objective of this analysis is to identify the most frequently used words in the image captions. It helps highlight the dominant terms in the descriptions and provides an overview of the vocabulary commonly used to describe the images in the dataset.

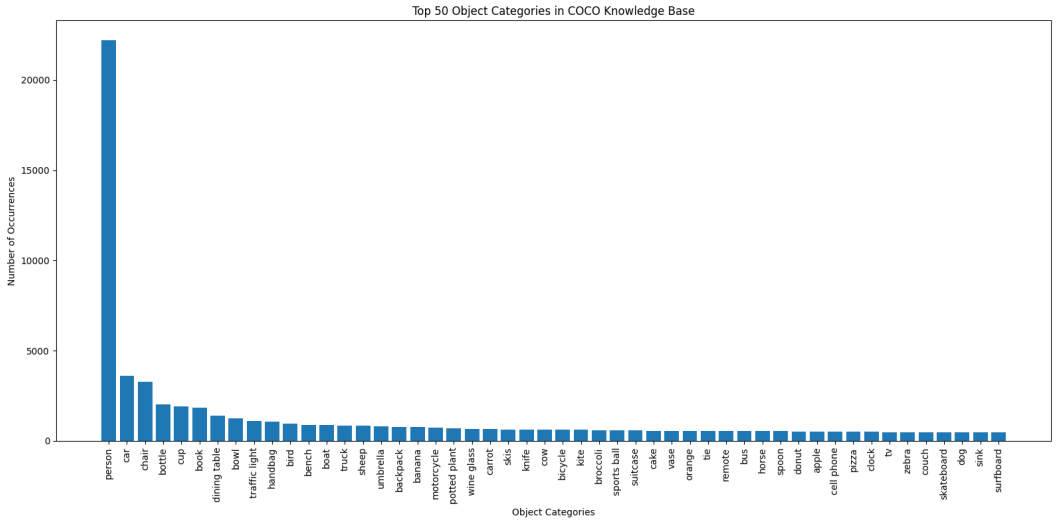


Figure 1.6: Most Frequent Words

1.5.6 Image Color Analysis

Dominant Color Extraction

The objective of this operation is to identify the dominant color in each image by determining the most frequent RGB color value. This analysis provides basic visual information about the images and helps characterize their overall color distribution. A sample of 500 images was selected for this analysis.

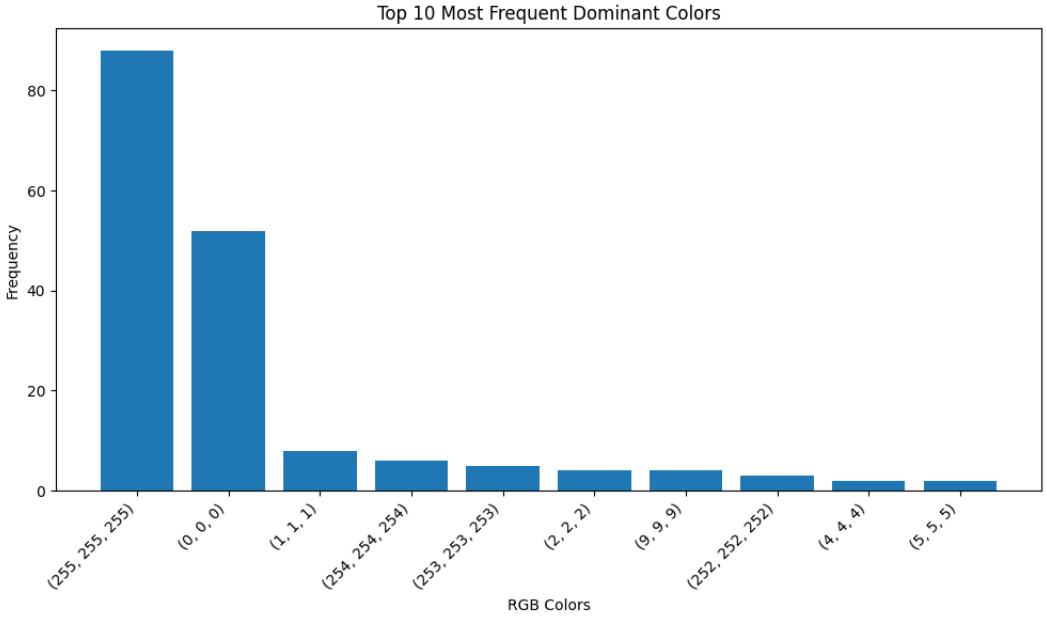


Figure 1.7: Most Frequent Words

1.6 Data Cleaning and Validation

1.6.1 Missing and Corrupted Images Detection

The objective of this step is to verify the integrity and availability of the images in the Knowledge Base. It identifies missing image files, detects corrupted images that cannot be properly verified, and distinguishes them from valid images. This step helps ensure that only accessible and readable images are used in the subsequent stages of the project.

Image Status	**Number of Images**
Valid Images	10,000
Missing Images	0
Corrupted Images	0
Total Images	**10,000**

Table 1.3: Image Integrity Verification Results

1.6.2 Image Format and Dimensions Extraction

The objective of this operation is to analyze the technical properties of the images in the Knowledge Base. It consists of identifying the image file formats, color modes, and dimensions "width and height". This analysis provides an overview of the image characteristics and helps identify variations in formats, color representations, and resolutions across the dataset.

1.6.3 Caption and Annotation Validation

The objective of caption cleaning is to improve the consistency of the textual descriptions by removing unnecessary variations and standardizing their format. Each caption is converted to lowercase, multiple spaces are replaced with a single space, punctuation and special characters are removed, and leading and trailing spaces are eliminated. The original captions are preserved, while the cleaned versions are stored separately for further processing.

1.7 Data Preprocessing and Splitting

1.7.1 Dataset Splitting

The objective of data splitting is to divide the dataset into three separate subsets: Knowledge Base (KB), Validation Set, and Test Set. Each subset has a specific role in the development and evaluation of the image captioning system. This separation

helps organize the data and ensures that the same images are not used across different subsets.

Dataset Subset	**Number of Images**	**Purpose**
Knowledge Base (KB)	10,000	Image retrieval
Validation Set	100	System validation
Test Set	100	Final system evaluation
Total	**10,200**	

Table 1.4: Dataset Splitting Summary

1.7.2 Split Organization

The image IDs belonging to each subset are stored in a JSON file named `splits_all.json`. This file contains five keys: `kb`, `val`, `test`, `test_strata`, and `note`.

The `kb`, `val`, and `test` keys store the image IDs assigned to the Knowledge Base, Validation Set, and Test Set, respectively. The `test_strata` key organizes the test image IDs into five categories according to their visual characteristics. The `note` key stores additional information about the split.

1.7.3 Test Reference Captions

In addition to the split information, a separate JSON file named `test_captions.json` is generated to store the reference captions associated with the test images. Each image ID is used as a key, and its corresponding captions are stored as a list of text descriptions. These reference captions will be used during the evaluation of the image captioning system.

1.7.4 Split Verification

Several checks were performed to verify the correctness of the dataset partitioning. These checks ensure that the three subsets have the expected sizes, contain no repeated image IDs, and remain independent of one another. Additional verification was performed on the Test Set stratification to confirm that the five categories contain exactly 100 unique image IDs and cover the complete Test Set without overlap.

1.7.5 Image Preprocessing

- **EXIF Orientation Correction:** Correcting image orientation using EXIF metadata to ensure that images are displayed in the correct direction.
- **RGB Color Conversion:** Converting all images to RGB color mode to ensure a consistent color representation.

- **JPEG Format Standardization:** Saving all images in JPEG format with a quality level of 95.
- **Processed Images Verification:** Checking the color modes and the total number of processed images.
- **Image Resizing:** Resizing images to a maximum size of 384×512 pixels while preserving their original aspect ratios, using the LANCZOS resampling method.
- **Resized Images Verification:** Checking the final image dimensions, file count, and compliance with the maximum size constraints.

1.7.6 Caption Preprocessing

- **Caption Extraction:** Extracting the original captions associated with the Knowledge Base (KB) images from the MS COCO annotations.
- **Text Normalization:** Converting captions to lowercase, removing punctuation and special characters, and eliminating extra whitespace.
- **Caption Cleaning:** Creating a cleaned version of each original caption while preserving its associated image ID.
- **Caption Dataset Storage:** Saving the original and cleaned captions in a CSV file named `kb_captions_clean.csv`.
- **Caption Audit:** Checking the number of captions per image, missing captions, and duplicate captions, then saving the audit results in `caption_audit.json`.

1.7.7 Data Organization and Metadata Preparation

The dataset information is organized into separate JSON and CSV files to facilitate data management and future processing.

- **Dataset Split Organization:** Storing the image IDs of the Knowledge Base, Validation Set, and Test Set in `splits_all.json`, along with the test image categories.
- **Caption Organization:** Storing image IDs, original captions, and cleaned captions in `kb_captions_clean.csv`.
- **Test Reference Captions:** Saving the reference captions associated with test images in `test_captions.json`.

- **Caption Audit Metadata:** Recording caption statistics, including the number of images, total captions, missing captions, and duplicates in `caption_audit.json`.

1.7.8 Data Leakage Verification

To prevent data leakage, several checks were performed to ensure the independence of the dataset subsets.

- **Dataset Size Verification:** Checking that the Knowledge Base contains 10,000 images, while the Validation and Test Sets each contain 100 images.
- **ID Overlap Verification:** Checking that there are no common image IDs between the Knowledge Base, Validation Set, and Test Set.
- **ID Uniqueness Verification:** Ensuring that each image ID appears only once within its corresponding subset.
- **Test Strata Verification:** Confirming that the five test categories contain exactly the same image IDs as the Test Set, without repetitions.

1.8 Embedding Generation and Vector Indexing

1.8.1 CLIP Model and Configuration

Images and captions are encoded with the pre-trained CLIP model `openai/clip-vit-base-patch32` (Hugging Face `transformers`), which maps both modalities into a shared 512-dimensional space. The model is used in inference mode only (no fine-tuning), on a Tesla T4 GPU in Google Colab.

1.8.2 Image Embedding Generation

Each preprocessed KB image (EXIF corrected, RGB) is passed through the CLIP processor and encoded with `get_image_features`, in batches of 32 with gradients disabled. Every batch is saved as a checkpoint (`.npz`) with its image IDs, so that the process can be resumed safely. One image (ID 3077) was detected as invalid (empty file) and excluded, leaving 9,999 image embeddings of shape $9{,}999 \times 512$.

1.8.3 Caption Embedding Generation

The cleaned captions (`caption_clean`) are encoded with the CLIP text encoder, in batches of 128 with a maximum length of 77 tokens, using the same checkpoint mechanism. This gives $50{,}024 \times 512$ embeddings, each linked to its image ID. The 5 captions of

the excluded image were removed, so the final caption corpus contains 50,019 captions.

1.8.4 Embedding Normalization and Storage

All embeddings are L2-normalized, so the inner product equals the cosine similarity. We verified that values are finite, norms equal 1 and image IDs are unique. Embeddings and their IDs are stored in NumPy format (`.npy`) for reproducibility.

1.8.5 FAISS Index Construction

The FAISS library (`faiss-cpu`) is installed in the Colab environment to be used as the vector database for similarity search over the L2-normalized image embeddings ($9{,}999 \times 512$) generated in the previous steps. The index itself is built in a later step of the pipeline.